

POLITECNICO DI TORINO

Master's Degree in Electronics



Master's Degree Thesis

Loss analysis of a 4-level, flying capacitor, bridge-less, totem-pole PFC

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MONTH 2026

Table of Contents

1	Simulations overview	3
1.1	Simulation setup	3
1.1.1	PFC simulation	3
1.2	Simulation results	3
	List of Tables	14
	List of Figures	15
	Bibliography	16

Chapter 1

Simulations overview

In order to perform the simulations a combination of SPICE and PLECS have been used. The first one was primarily used to validate all the device models of the single components, and produce all the necessary data to be given to the PLECS simulator, in particular the energy-on, energy-off of the MOSFETs in unision with their driver circuit.

1.1 Simulation setup

1.1.1 PFC simulation

The entire PFC system was simulated using the PLECS software, this is much faster than any SPICE simulation, since PLECS runs on a system level, and the simulated devices are either on or off, no transients are present, that's why the energy-on/energy-off SPICE simulations were fundamental, since those data are needed by the PLECS software to perform any kind of loss analysis.

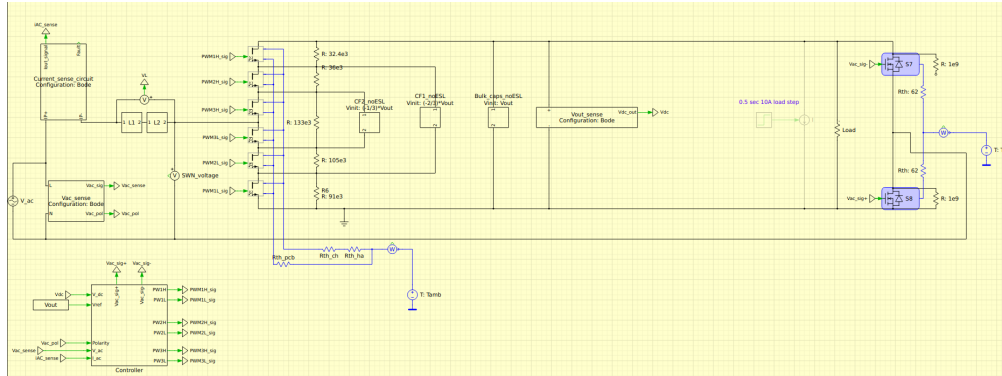


Figure 1.1: PFC PLECS schematich

1.2 Simulation results

These simulations DO NOT take into account:

- House-keeping PSU stand-by power and efficiency
- Gate driver consumption
- Sensing circuits
- Controller consumption

Only the power section components have been included with some limitations:

- NO ESL on Flying caps
- NO ESL on HF decoupling caps
- NO ESL on Electrolytic bulk caps
- NO AC losses models for inductors

All remaining elements have been modeled as closely as possible to the EPC91107 datasheet

- feedback filtering chain (low pass filters as transfer functions)
- controller to gate driver filtering chain(included in GaN/Si energy calculations)
- devices parasitic
- 230V 50Hz as AC input
- 400V as DC output (variable current load)

The PLECS software emulate the system behaviour for a time span of 1.3 seconds.

All the losses analysis has been computed over an average of the last 5 periods of the 50Hz cycle of the simulation.

The system has been initialized with some initial condition in such a way that all electrical quantities can be considered in a steady-state region.

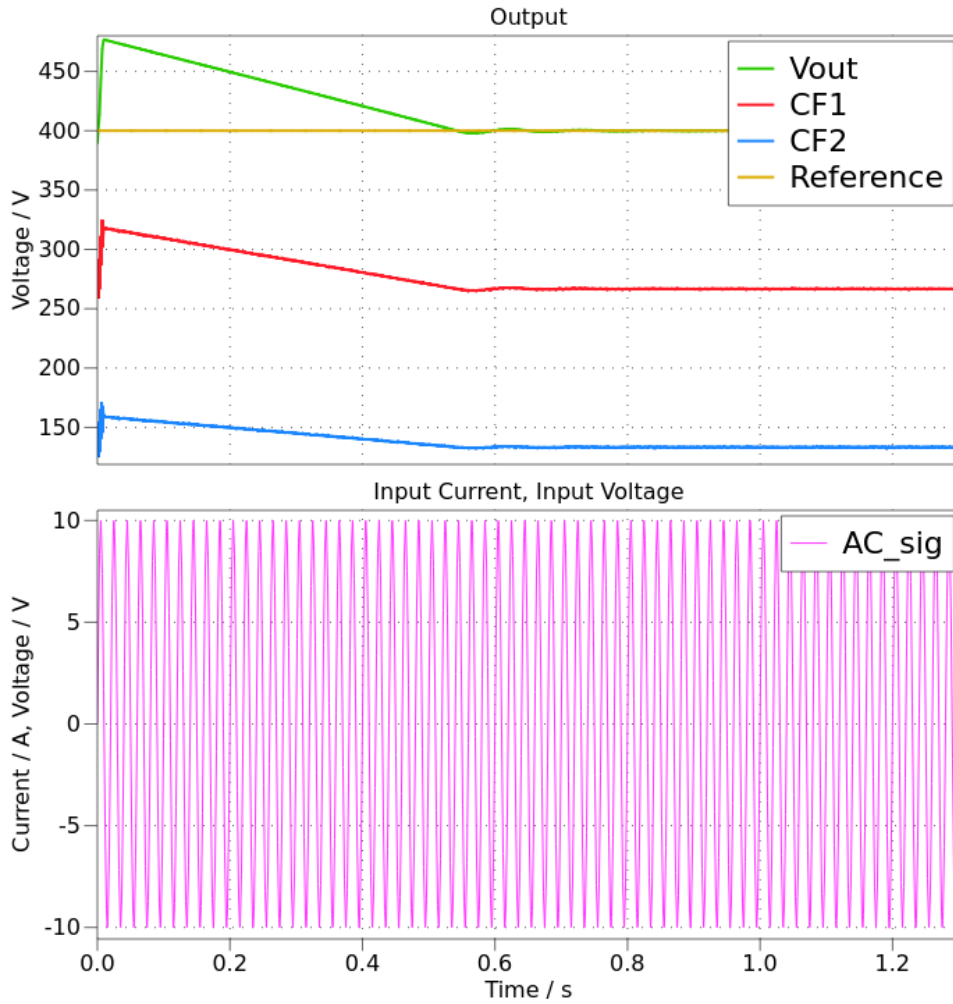


Figure 1.2: Voltage during simulations

The same reasoning can be made for the thermal initial condition, but a piecewise finer approach has been taken in order to get better results, since thermal behaviour has time constants greater than the simulation time. The simulation was considered at equilibrium when during the entire simulation time the junction temperature of the active devices was somewhat constant.

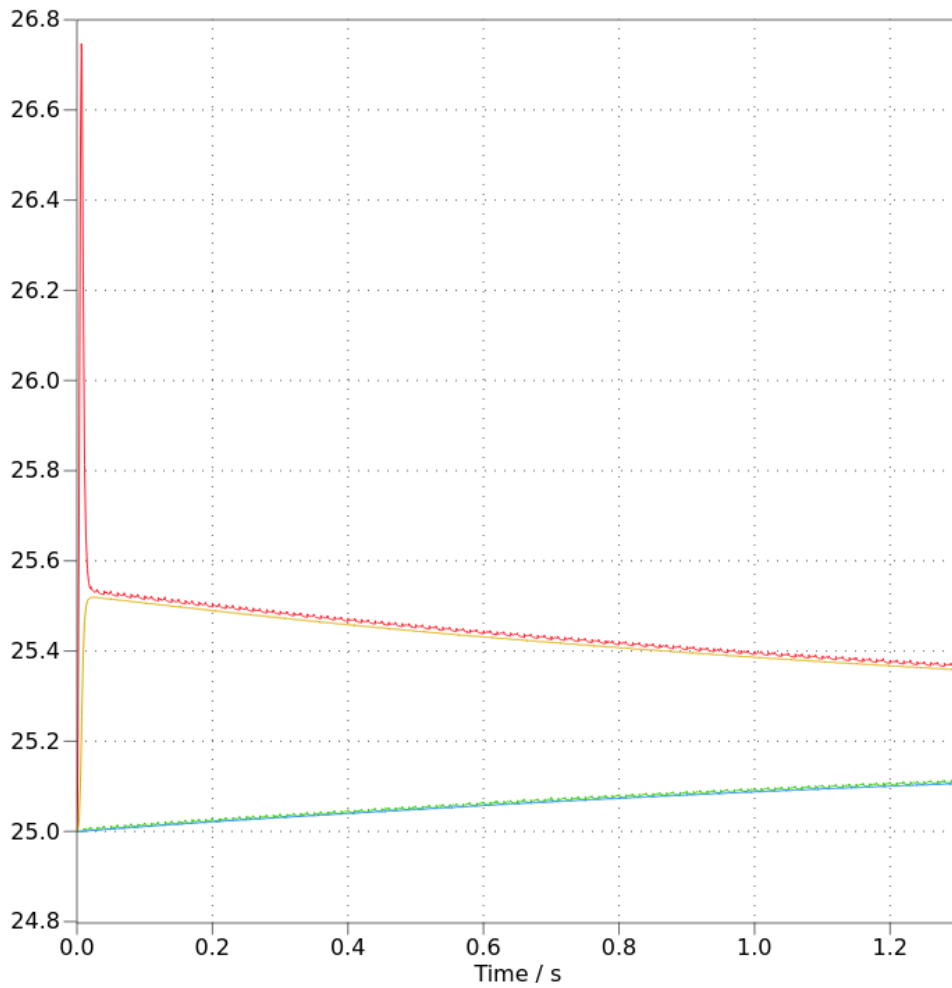


Figure 1.3: Example of stable temperature

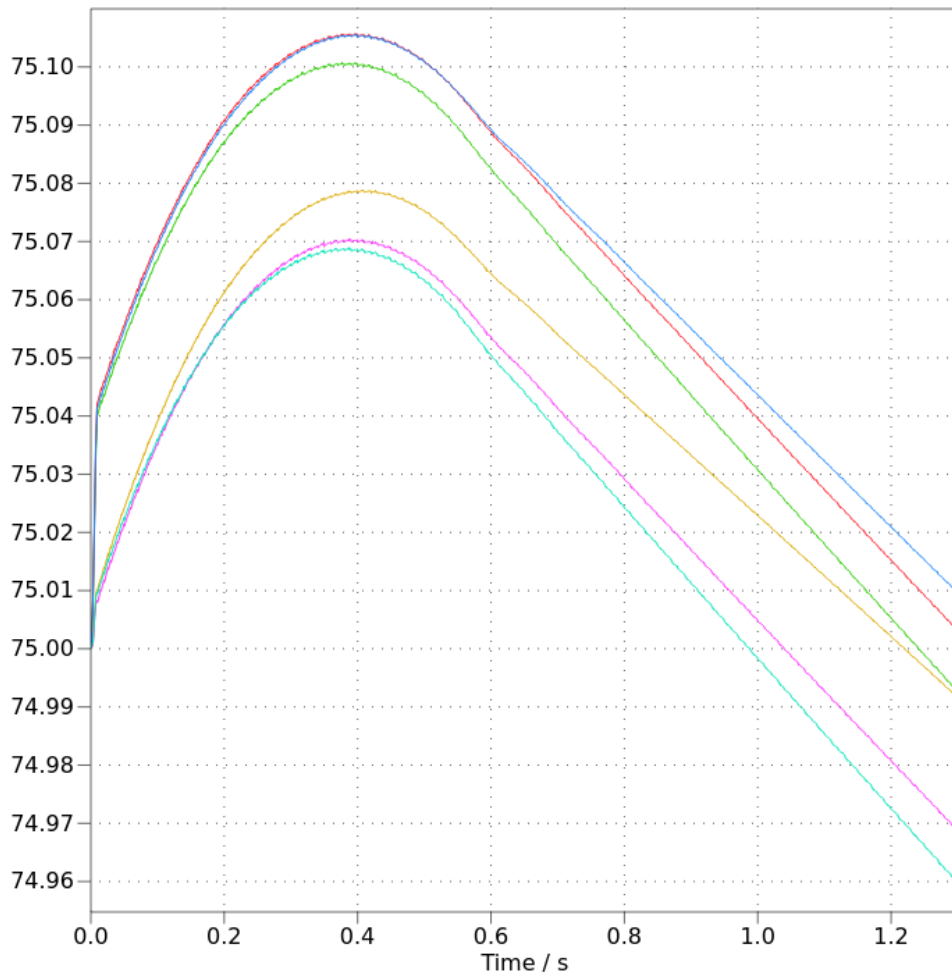


Figure 1.4: Example of NOT stable temperature

In order to verify that the simulated system has similar behaviour to the reality, a 5kW simulation has been performed and the waveform are analyzed and compared to the 91107 datasheet [1]

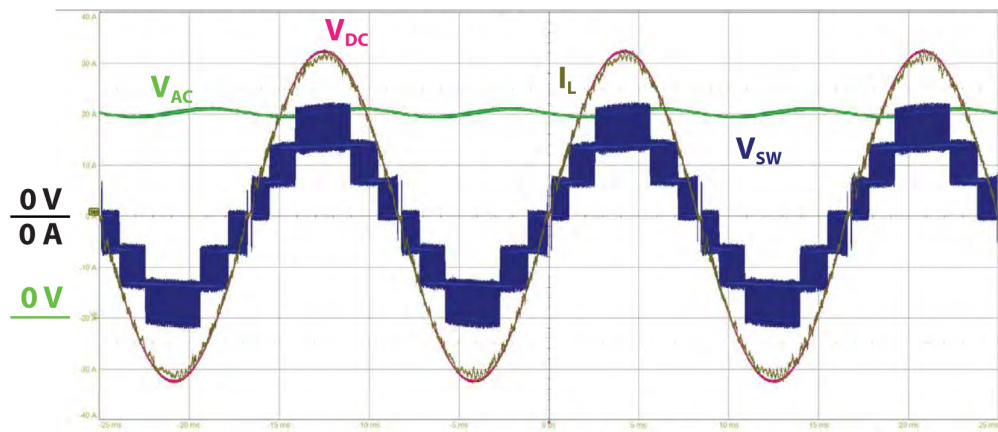


Figure 1.5: 5kW test

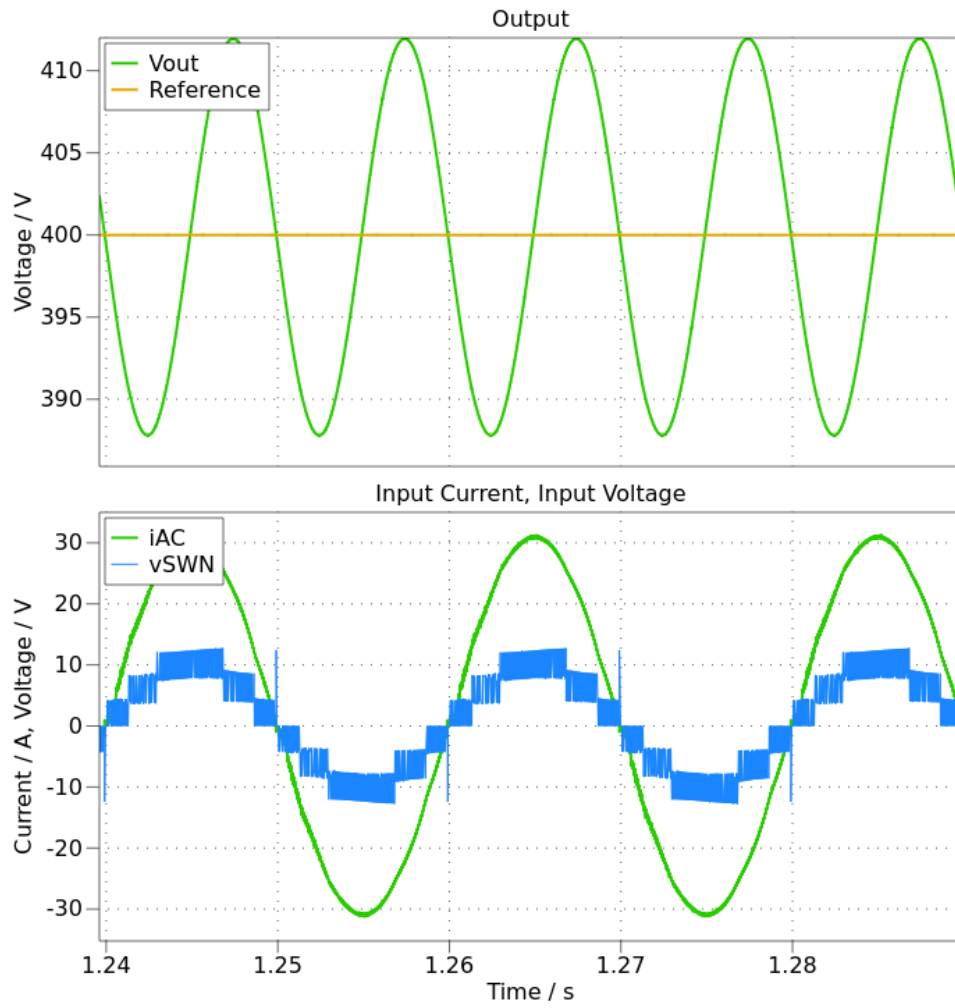


Figure 1.6: 5kW sims

A swept power analysis have been performed from 50W to 5050W.

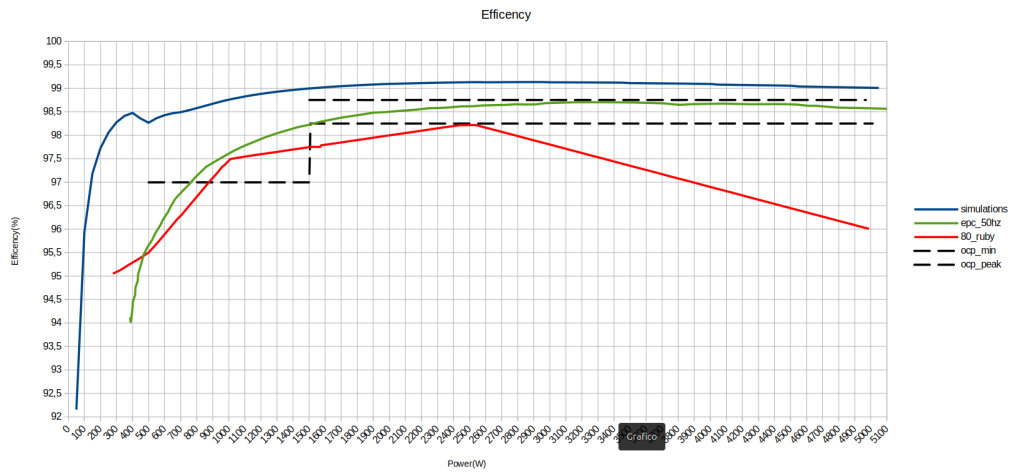


Figure 1.7: Efficiency in %

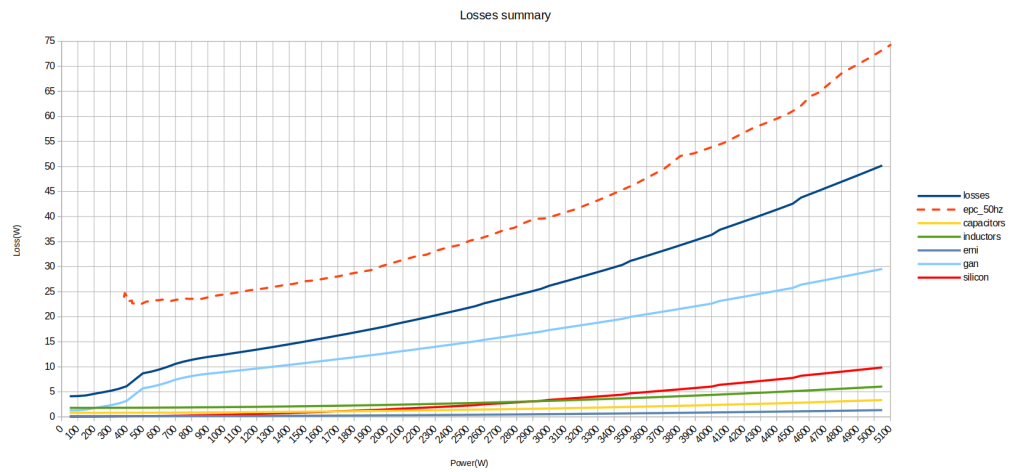


Figure 1.8: Losses summary

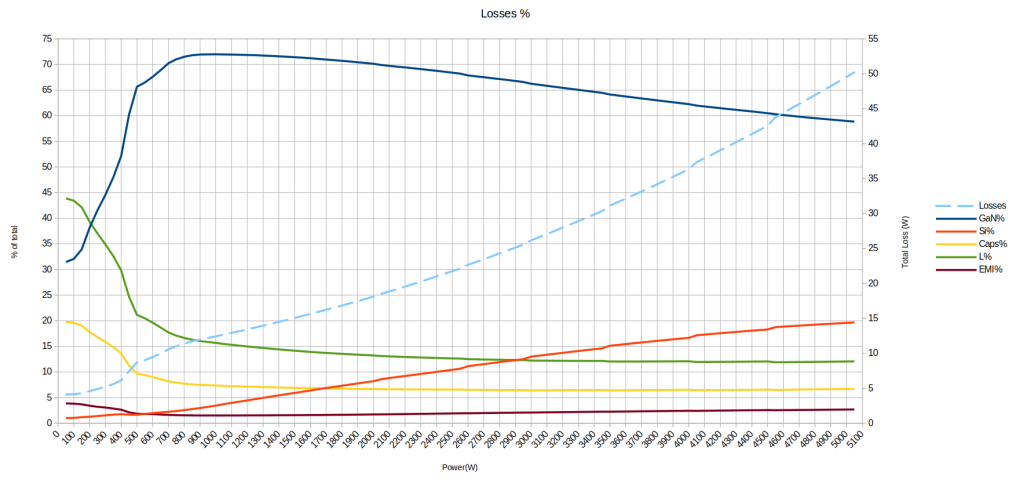


Figure 1.9: Losses % of total

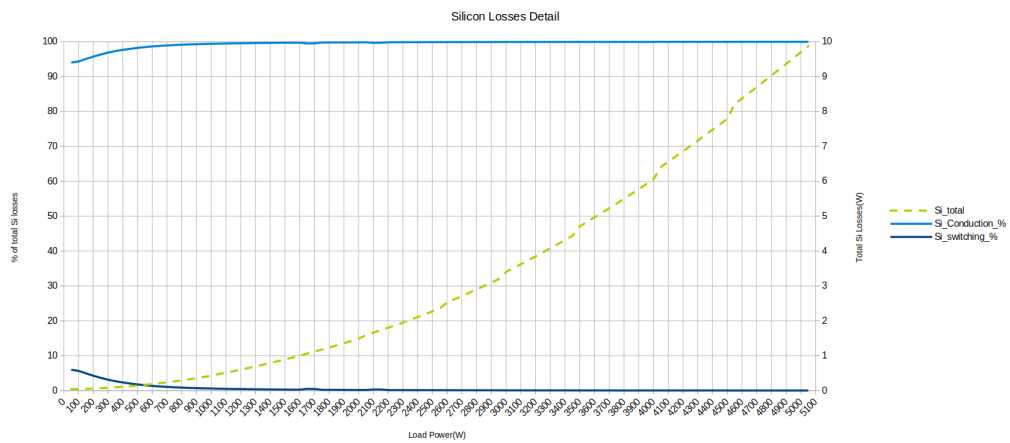


Figure 1.10: Si losses detail

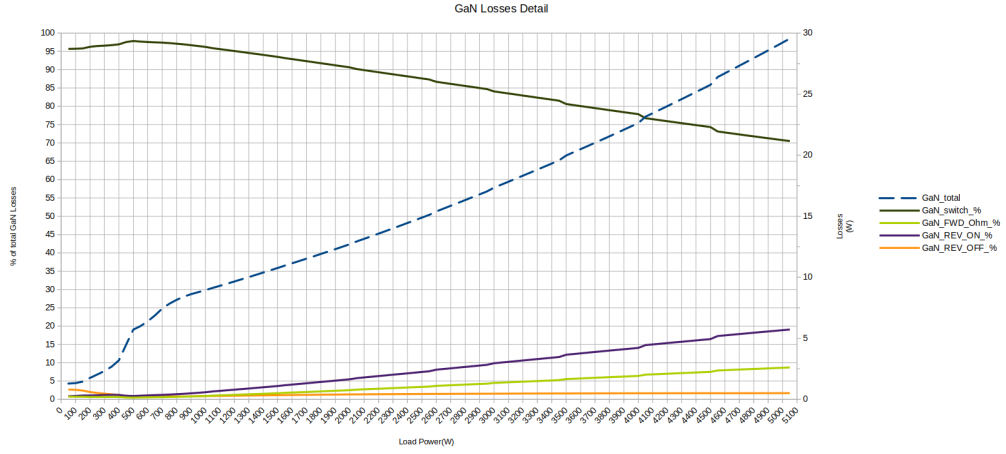


Figure 1.11: GaN losses detail

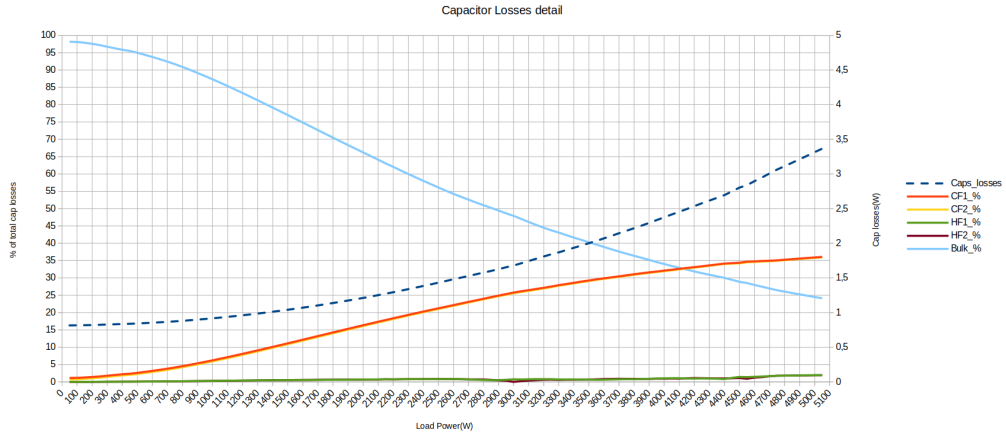


Figure 1.12: Capacitors losses detail

It must be considered tho that simulations results are still not perfectly fitting the real device behaviour, some aspects of the simulations have yet to be measured and so modeled, this leads to a discrepancy, which is highlited down below.

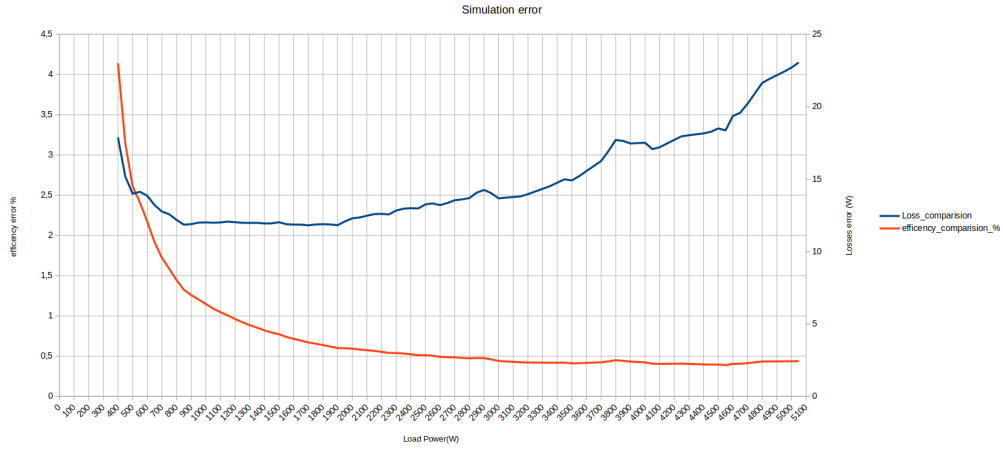


Figure 1.13: Simulation error

A crude thermal analysis of the system has been performed, however since there is no accurate model of the real thermal chain, all of these results are based on an educated guess, in order to tune up these values a real world measurement of the active devices temperature must be performed.

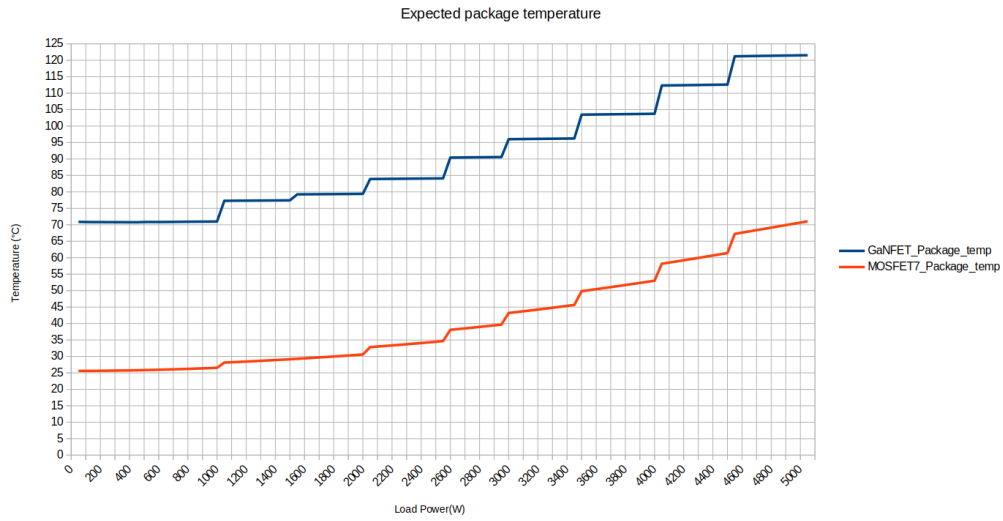


Figure 1.14: Active devices expected package temperatures

List of Tables

List of Figures

1.1	PFC PLECS schematich	3
1.2	Voltage during simulations	5
1.3	Example of stable temperature	6
1.4	Example of NOT stable temperature	7
1.5	5kW test	8
1.6	5kW sims	9
1.7	Efficiency in %	10
1.8	Losses summary	10
1.9	Losses % of total	11
1.10	Si losses detail	11
1.11	GaN losses detail	12
1.12	Capacitors losses detail	12
1.13	Simulation error	13
1.14	Active devices expected package temperatures	13

Bibliography

- [1] EPC-CO. *Evaluation Kit EPC91107KIT Quick Start Guide*. https://epc-co.com/epc/portals/0/epc/documents/guides/EPC91107_qsg.pdf. 2025 (cit. on p. 7).